# Pancreatitis with ARDS



## Section I: Scenario Demographics

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|---|---|
| Scenario Title: | Pancreatitis with ARDS |
| Date of Development: | 29/06/2015 (DD/MM/YYYY) |
| Target Learning Group: | ☐ Juniors (PGY 1 – 2) ☐ Seniors (PGY ≥ 3) ☒ All Groups |

## Section II: Scenario Developers



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|---|---|
| Scenario Developer(s): | Kyla Caners |
| Affiliations/Institution(s): | McMaster University |
| Contact E-mail (optional): | kcaners@gmail.com |



## Section III: Curriculum Integration

| Learning Goals & Objectives | |
|---|---|
| Educational Goal: | To enhance resuscitation and team management skills. |
| CRM Objectives: | Communicate effectively with team members in the care of a complex, critically ill patient. |
| Medical Objectives: | 1) Begin broad diagnostic workup while simultaneously resuscitating a critically ill patient.<br>2) Perform aggressive fluid resuscitation and identify when vasopressor required.<br>3) Recognize onset of ARDS and demonstrate safe approach to intubating hypotensive patient with ARDS. |

| Case Summary: Brief Summary of Case Progression and Major Events |
|---|
| A 50 year-old female who was "on a bender" over the weekend now presents with diffuse abdominal pain and persistent nausea and vomiting. She will have a diffusely tender abdomen, a BP of 80/40, and be tachycardic. The team will need to work through a broad differential diagnosis and should fluid resuscitate aggressively. Once the patient has received 6L of fluid, she will become tachypneic and hypoxic and require intubation. The team will be given a lipase result just prior. |



| References |
|---|
| Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby. |
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## Section IV: Scenario Script

| A. Scenario Cast & Realism | | | |
|---|---|---|---|
| Patient: | ☒ Computerized Mannequin | Realism: | ☒ Conceptual |
| | ☐ Mannequin | | ☐ Physical |
| | ☐ Standardized Patient | *Select most important dimension(s)* | ☐ Emotional/Experiential |
| | ☐ Hybrid | | ☐ Other: |
| | ☐ Task Trainer | | ☐ N/A |

| Confederates | Brief Description of Role |
|---|---|
| None required. | |

| B. Required Monitors | | |
|---|---|---|
| ☒ EKG Leads/Wires | ☐ Temperature Probe | ☐ Central Venous Line |
| ☒ NIBP Cuff | ☐ Defibrillator Pads | ☐ Capnography |
| ☒ Pulse Oximeter | ☒ Arterial Line | ☐ Other: |

| C. Required Equipment | | |
|---|---|---|
| ☒ Gloves | ☒ Nasal Prongs | ☐ Scalpel |
| ☒ Stethoscope | ☒ Venturi Mask | ☐ Tube Thoracostomy Kit |
| ☐ Defibrillator | ☒ Non-Rebreather Mask | ☐ Cricothyroidotomy Kit |
| ☒ IV Bags/Lines | ☒ Bag Valve Mask | ☐ Thoracotomy Kit |
| ☒ IV Push Medications | ☒ Laryngoscope | ☐ Central Line Kit |
| ☐ PO Tabs | ☐ Video Assisted Laryngoscope | ☐ Arterial Line Kit |
| ☐ Blood Products | ☒ ET Tubes | ☐ Other: |
| ☐ Intraosseous Set-up | ☐ LMA | ☐ Other: |

| D. Moulage |
|---|
| None required. (Could choose to put a paper bag with empty beer bottle in it beside patient.) |

| E. Approximate Timing | | | | | |
|---|---|---|---|---|---|
| Set-Up: | 3 min | Scenario: | 12 min | Debriefing: | 15 min |

## Section V: Patient Data and Baseline State

| A. Clinical Vignette: To Read Aloud at Beginning of Case |
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| Patricia is a 50 year old female who presents with epigastric abdominal pain. It's been persistent for the last 24 hours and radiates through to her back. She has been nauseous all day and has been vomiting so much she "can't keep anything down." She was "on a bender" this weekend drinking beer and whiskey. |

| B. Patient Profile and History | | | | |
|---|---|---|---|---|
| Patient Name: Patricia Abrams | | Age: 50 | | Weight: 60kg |
| Gender: ☐ M ☒ F | | Code Status: Full | | |
| Chief Complaint: Abdominal pain | | | | |
| History of Presenting Illness: Has been on a bender for the last few days. Now having worsening epigastric abdominal pain and persistent nausea and vomiting. Unable to keep anything down. | | | | |
| Past Medical History: | **EtOH abuse** | Medications: | HCTZ 25mg daily – doesn't take | |
| | **HTN** | | | |
| | | | | |
| Allergies: sulfa – rash | | | | |
| Social History: **Smokers 1ppd. Drinks heavily on weekends in binges. Has never had withdrawal.** | | | | |

| Review of Systems: | | | | |
|---|---|---|---|---|
| | CNS: | **Feels dizzy and weak when stands.** | | |
| | HEENT: | Nil. | | |
| | CVS: | No CP. | | |
| | RESP: | Mild SOB. | | |
| | GI: | **Epigastric umbilical pain that bores into her back. Nausea with emesis all day. "Can't keep anything down." Loose BM x1 this am.** | | |
| | GU: | Nil. | | |
| | MSK: | Nil. | INT: | Nil. |

| C. Baseline Simulator State and Physical Exam | | | |
|---|---|---|---|
| ☐ No Monitor Display | ☒ Monitor On, no data displayed | ☐ Monitor on Standard Display | |
| HR: 125/min | BP: 80/40 | RR: 20/min | $O_2$SAT: 93% RA |
| Rhythm: Sinus tach | T: 35.9°C | Glucose: 5.8 mmol/L | GCS: 15 (E4 V5 M6) |
| General Status: Looks unwell. | | | |

| | | | |
|---|---|---|---|
| CNS: | Alert and oriented. | | |
| HEENT: | PERLA, 3mm, normal EOM. No signs HI. | | |
| CVS: | Normal, no murmur. | | |
| RESP: | GAEB. **Fine crackles at bases.** Mildly tachypneic. | | |
| ABDO: | **Soft, but diffusely tender. Maximal tenderness over epigastrum.** | | |
| GU: | Nil. | | |
| MSK: | Nil. | SKIN: | Nil. **(No jaundice.)** |



## Section VI: Scenario Progression

| Scenario States, Modifiers and Triggers | | | |
|---|---|---|---|
| Patient State | Patient Status | Learner Actions, Modifiers & Triggers to Move to Next State | |
| **1. Baseline State**<br>Rhythm: Sinus tach<br>HR: 125/min<br>BP: 80/40<br>RR: 20/min<br>$O_2$SAT: 93% RA<br>T: 35.9°C | Alert. Complaining of diffuse abdo pain. Retching. | <u>Learner Actions</u><br>-☐ Monitors, apply oxygen<br>-☐ 2 IVs, start NS bolus x2L<br>-☐ Blood work (abdo/cardiac)<br>-☐ Analgesia & anti-emetic<br>-☐ ECG<br>-☐ Upright AXR – R/o free air<br>-☐ Cap **sugar – 5.8**<br>-☐ Broad spectrum antibiotics<br>-☐ Bedside U/S r/o FF or AAA | <u>Modifiers</u><br>*Changes to patient condition based on learner action*<br>- $O_2$ applied → sats to 95%<br>- Analgesia → HR to 120<br>- Anti-emetic → stops retching<br><br><u>Triggers</u><br>*For progression to next state*<br>- 2L bolus or 5 min → **2. Fluid non-responsive** |
| **2. Fluid Non-Responsive**<br><br>BP → 70/30<br>HR → 130 | Patient still alert, but less talkative. Groaning a little. | <u>Learner Actions</u><br>-☐ Bedside U/S (if not yet done)<br>-☐ Give 2L NS again (4L total)<br>-☐ Start vasopressor | <u>Modifiers</u><br>- 4L IVF total → BP 85/45<br><br><u>Triggers</u><br>- 4-6L IVF given → **3. Hypoxic**<br>- 8 min → **3. Hypoxic** |
| **3. Hypoxic**<br><br>$O_2$SAT → slow drop to 85%<br>RR → 24 | ****Start state by giving lipase result****<br>Diffuse crackles. Increased WOB. | <u>Learner Actions</u><br>-☐ Apply $O_2$ or increase $O_2$<br>-☐ Prepare for intubation<br>-☐ BVM when $O_2$ not improving (add PEEP/BiPap)<br>-☐ CXR<br>-☐ Prepare for intubation | <u>Modifiers</u><br>- BVM → $O_2$SAT 88%<br>- Bipap or BVM with PEEP valve → $O_2$SAT 92%<br><br><u>Triggers</u><br>- Intubation → **4. Intubation** |
| **4. Intubation**<br><br>Vitals as per previous state | | <u>Learner Actions</u><br>-☐ Apneic oxygenation<br>-☐ Consider Bipap as bridge to intubation (or BVM with PEEP)<br>-☐ Ketamine for induction<br>-☐ Vasopressor at bedside<br>-☐ Administer paralytic | <u>Modifiers</u><br>- Bipap or BVM with PEEP valve → $O_2$SAT 92%<br><br><u>Triggers</u><br>- Successful intubation → **4. Resolution** |
| **4. Resolution**<br><br>$O_2$SAT → 94%<br>RR → 12 (Vented)<br>HR → 110<br>BP → 85/45 | Intubated & sedated. | <u>Learner Actions</u><br>-☐ Call ICU<br>-☐ Post-intubation CXR<br>-☐ Lung-protective vent settings (TV 6-8ml/kg)<br>-☐ Insert foley<br>-☐ Insert OG<br>-☐ Titrate vasopressors<br>-☐ Start sedation | **ICU arrives → END CASE** |



## Section VII: Supporting Documents, Laboratory Results, & Multimedia

| Laboratory Results | | | | | | |
|---|---|---|---|---|---|---|
| **Na: 131** | **K: 3.1** | **Cl: 100** | **$HCO_3$: 8** | **BUN: 10** | **Cr: 132** | **Glu: 13** |

| | | | | | |
|---|---|---|---|---|---|
| VBG | **pH: 7.15** | **$PCO_2$: 26** | **$PO_2$: 45** | **$HCO_3$: 8** | **Lactate: 6** |

| | | | |
|---|---|---|---|
| **WBC: 22** | Hg: 110 | Hct: 0.27 | **Plt: 145** |
| TB: 12 | GGT: 50 | ALP: 26 | AST: 45 |

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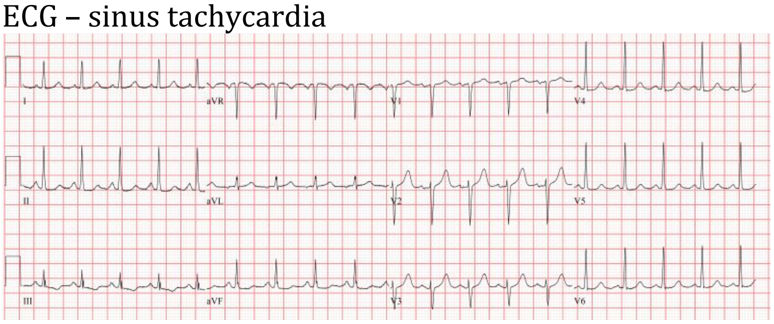
|---|---|
| **Lipase 1000** | **INR 1.4** |

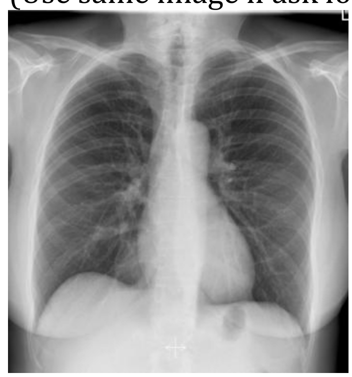
### Images (ECGs, CXRs, etc.)

ECG – sinus tachycardia



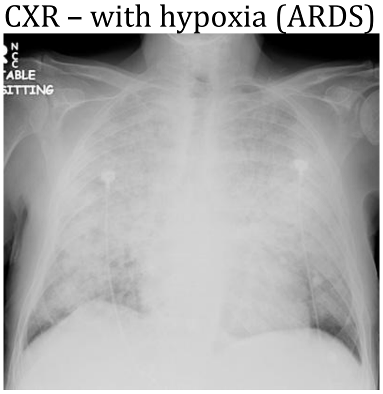
ECG source: http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/12/sinus-tachycardia.jpg

CXR – initial (normal)
(Use same image if ask for AXR for free air)



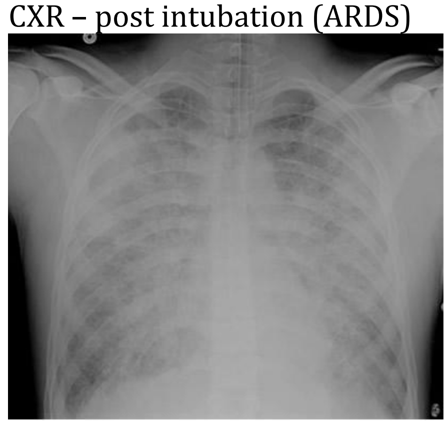
Source: http://radiopaedia.org/articles/normal-position-of-diaphragms-on-chest-radiography

CXR – with hypoxia (ARDS)



CXR source: http://www.radiology.vcu.edu/programs/residents/quiz/pulm_cotw/PulmonConf/09-03-04/68yM%2008-03-04%20CXR.jpg

CXR – post intubation (ARDS)



CXR source: http://courses.washington.edu/med620/images/mv_c3fig1.jpg

### Ultrasound Video Files (if applicable)

| | |
|---|---|
| U/S abdomen – normal (no FF) | U/S aorta – normal (no AAA) |
| Pericardial U/S – normal (no effusion) | |



## Section VIII: Debriefing Guide

| General Debriefing Plan | | | |
|---|---|---|---|
| ☐ Individual | ☒ Group | ☐ With Video | ☒ Without Video |

| Objectives | |
|---|---|
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**Sample Questions for Debriefing**

1) What was on your differential diagnosis for this patient? What was your thought process when the patient's blood pressure wasn't responding to fluids?
2) Why do you think the patient became hypoxic? Does that fit with the rest of the clinical picture?
3) How did you attempt to optimize your patient's oxygenation before intubation? Is there anything else you could have done in this clinical context?
4) Do you feel that roles were allocated well in this scenario? Was it clear who was managing what?

| Key Moments |
|---|
| Recognition of toxic patient |
| Recognition of fluid non-responsiveness |
| Management of hypoxia |